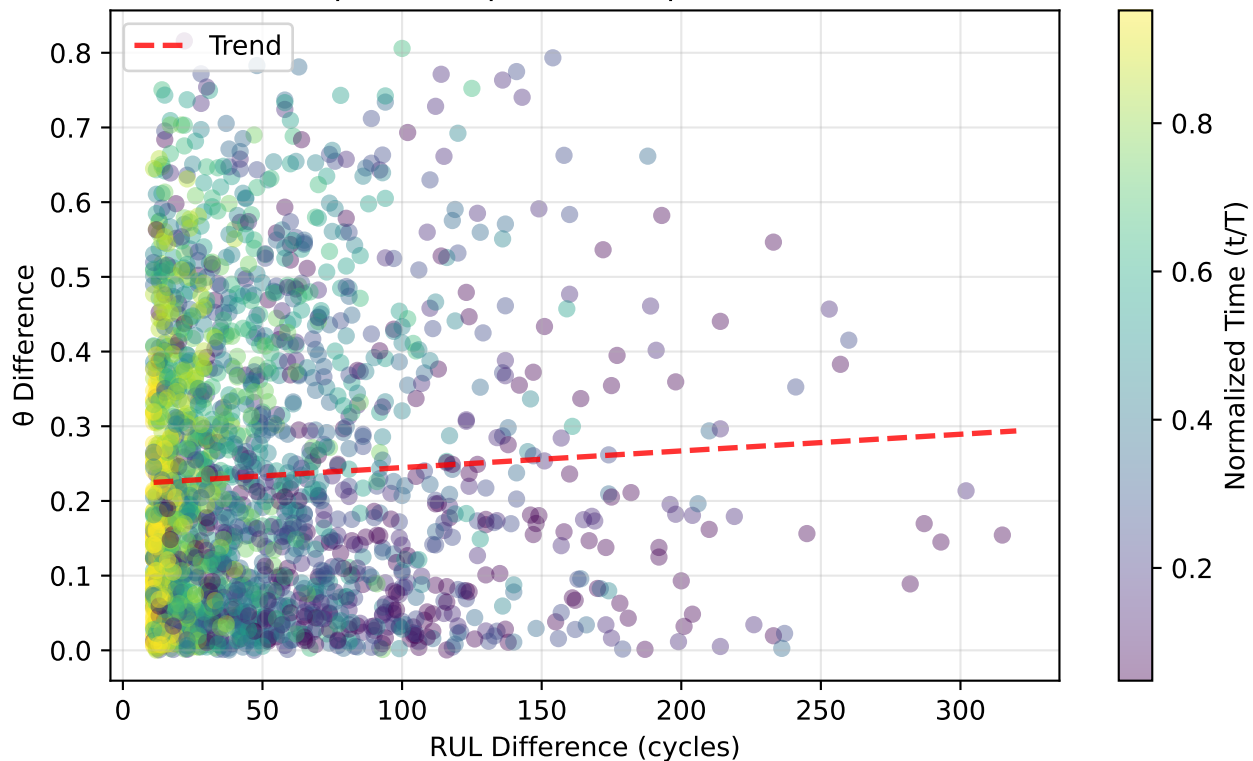
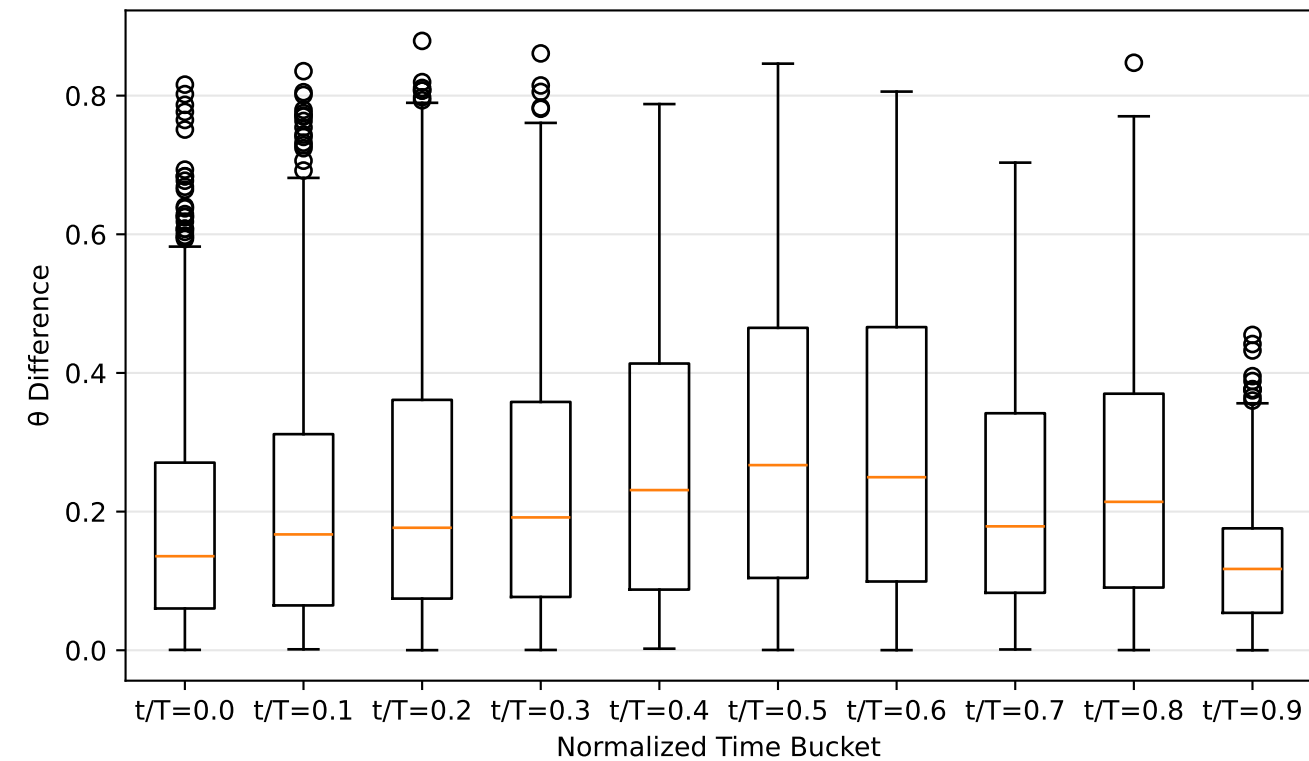


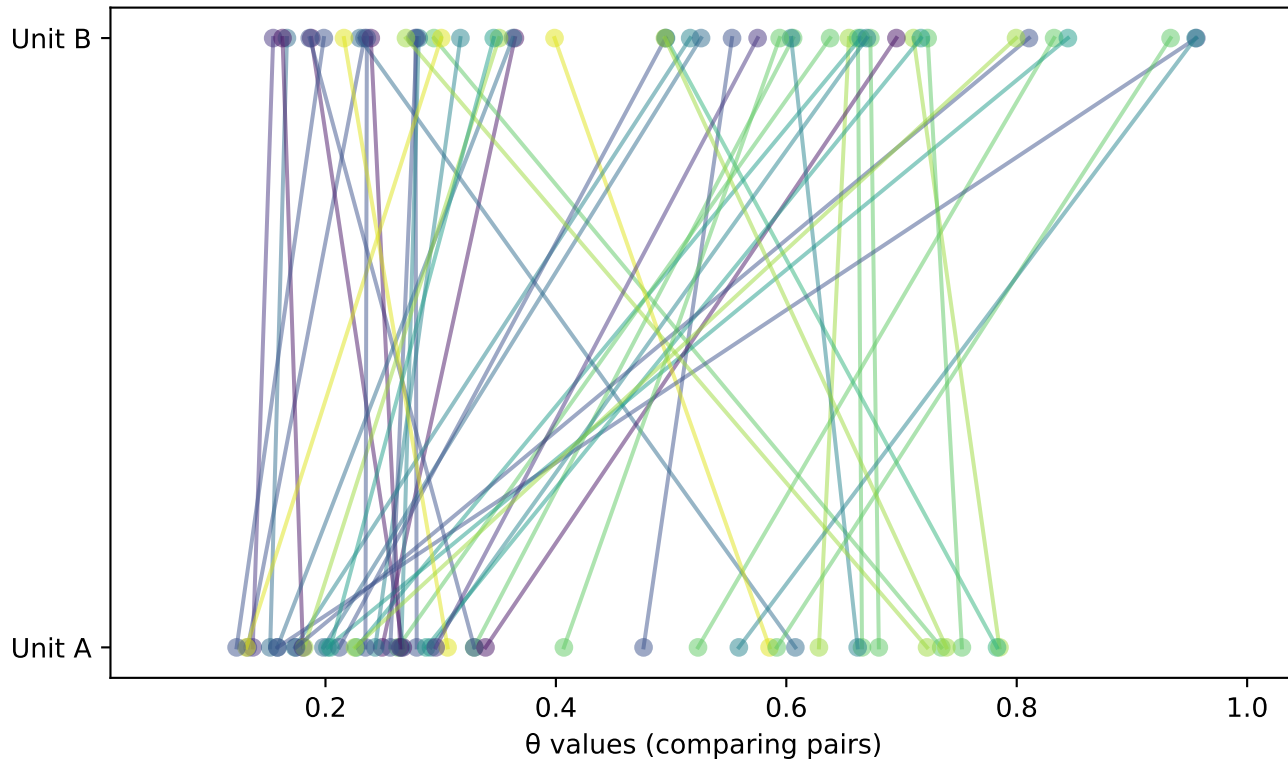
Health State Difference vs θ Difference
(Spearman $\rho=0.0782$, $p=3.04e-08$)



θ Difference Across Different Times



Sample Pairs: Showing θ Differences Within Pairs



Test 1c: Same-Time Different-Health Analysis

Dataset: FD004
Latent Dim: 4
Window Size: 10

Results:

Comparison Pairs: 5000

RUL Difference:
Mean: 48.4 cycles
Median: 33.0 cycles
Max: 320.0 cycles

θ Difference:
Mean: 0.2332
Median: 0.1793
Max: 0.8789

Correlation:
Spearman $\rho = 0.0782$
 p -value = $3.04e-08$

Interpretation:

If $\rho \gg 0$ and $p < 0.05$:
→ θ TRACKS STATE (different health leads to different θ even at same t/T)

If $\rho \approx 0$ or $p > 0.05$:
→ θ may be learning just t/T (health difference doesn't affect θ)